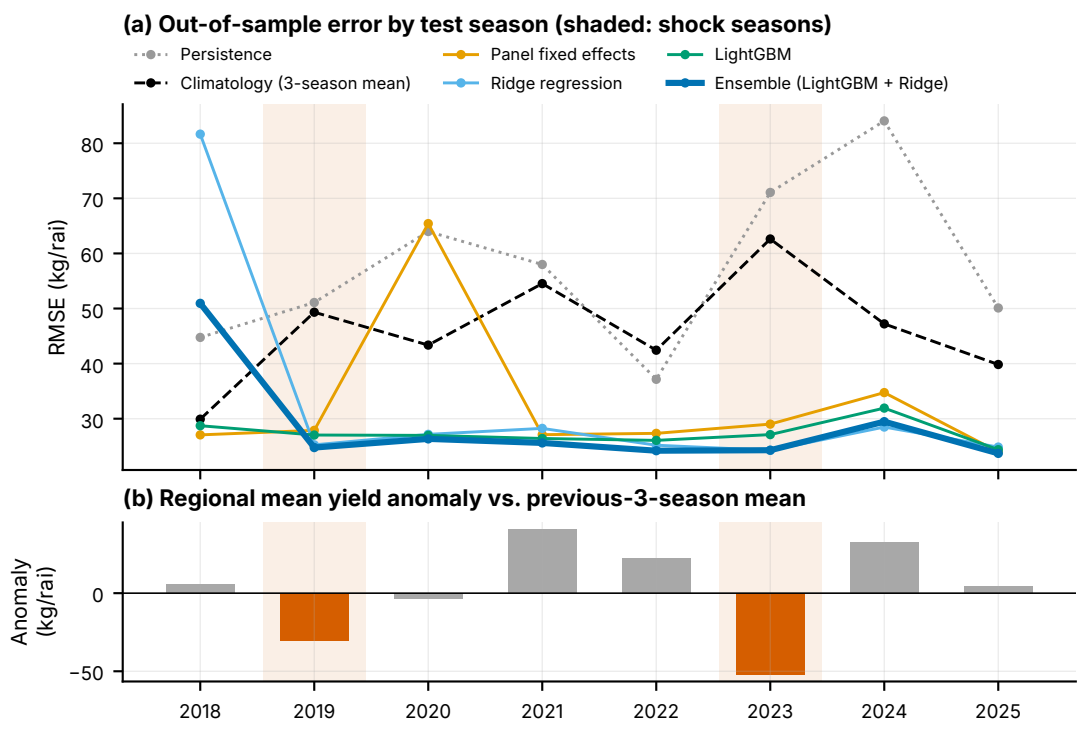
(a) Out-of-sample error by test season (shaded: shock seasons)
Persistence
Climatology (3-season mean)
Panel fixed effects
Ridge regression
LightGBM
Ensemble (LightGBM + Ridge)
RMSE (kg/rai)
80
70
60
50
40
30
(b) Regional mean yield anomaly vs. previous-3-season mean
Anomaly (kg/rai)
0
−50
2018
2019
2020
2021
2022
2023
2024
2025